

THE NEW SCHOOL • EUGENE LANG COLLEGE OF LIBERAL ARTS

Day 02

When Does a Sum Become an Area?

LMTH 2040 — Calculus: A Multidisciplinary Approach

Monday, August 31, 2026

Key Idea

Today we will begin with a pattern you already know and ask what happens when we change its scale. Our path will be

pattern $\longrightarrow$ sum $\longrightarrow$ picture $\longrightarrow$ computation $\longrightarrow$ area.

The central question is:

When does a sum become an area?

1. Triangular Numbers Revisited

Recall the triangular numbers

$$T_n = 1 + 2 + 3 + \cdots + n.$$

For example,

$$T_1 = 1, \quad T_2 = 3, \quad T_3 = 6, \quad T_4 = 10.$$

Compare

With your group, compare your solutions to the triangular-number problem from Homework 01.

Identify at least two different ways of representing or determining T_n .

Representation 1:

Representation 2:

What does each representation reveal?

Complete the table.

	What is easy to see?	What is difficult to see?
Picture / diagram		
Sum		
Formula		
Computation		

Reflection

A numerical pattern can provide *evidence* for a conjecture. A mathematical argument explains why the conjecture must continue to hold.

Where does this distinction appear in your solution to the triangular number problem?

2. Changing the Scale

Suppose we place the n th triangular number inside an $n \times n$ square.

The square contains n^2 unit squares.

The triangle contains T_n objects.

So consider the ratio

$$A_n = \frac{T_n}{n^2}.$$

Calculate

Without using the general formula for T_n , calculate:

n	T_n	$A_n = T_n/n^2$
1		
2		
3		
4		
5		
10		

Conjecture

What appears to happen to A_n as n becomes large?

Use the formula

Now use

$$T_n = \frac{n(n+1)}{2}$$

to simplify

$$A_n = \frac{T_n}{n^2}.$$

What does your expression suggest happens as n becomes very large?

Reflection

We have changed the question.

Instead of asking how large T_n becomes, we are asking how large it is *relative to the square containing it*.

Why might changing the scale reveal a pattern that is difficult to see in the original numbers?

3. Turn the Sum into a Picture

Start again with

$$A_n = \frac{T_n}{n^2} = \frac{1}{n^2} (1 + 2 + \cdots + n).$$

Rewrite this expression by distributing the two factors of n :

$$A_n = \frac{1}{n} \left(\frac{1}{n} + \frac{2}{n} + \frac{3}{n} + \cdots + \frac{n}{n} \right).$$

A geometric interpretation

Consider a square with side length 1.

Divide the horizontal axis into n equal pieces.

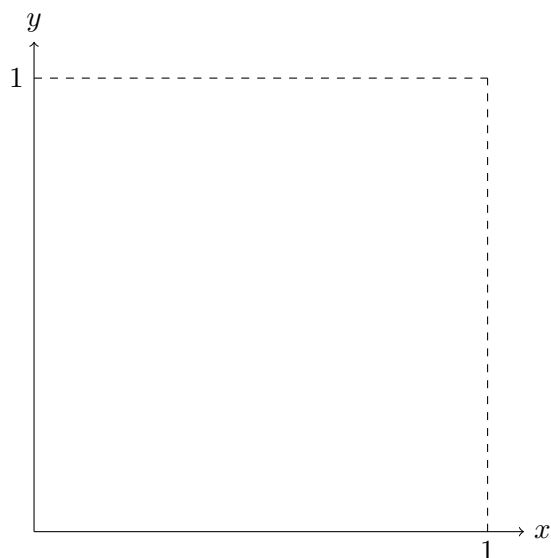
Each piece therefore has width



Now interpret

$$\frac{1}{n}, \quad \frac{2}{n}, \quad \frac{3}{n}, \quad \dots, \quad \frac{n}{n}$$

as heights.



For $n = 4$, draw four rectangles with width $1/4$ and heights

$$\frac{1}{4}, \quad \frac{2}{4}, \quad \frac{3}{4}, \quad \frac{4}{4}.$$

Area

What is the area of the first rectangle?

$$\text{area} = \boxed{} \cdot \boxed{} = \boxed{}$$

What is the area of the second?

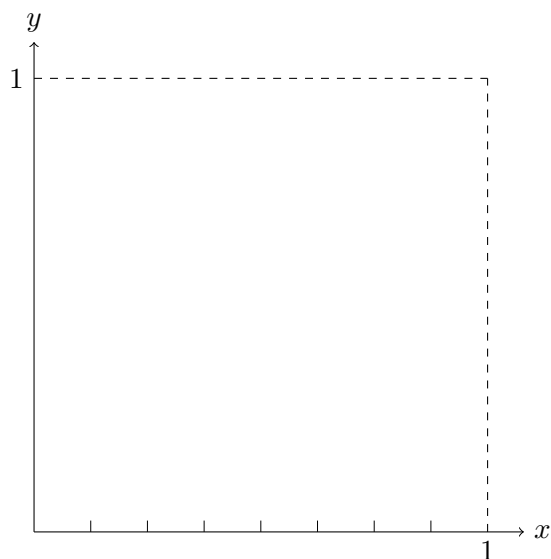
$$\text{area} = \boxed{} \cdot \boxed{} = \boxed{}$$

What is the total area of all four rectangles?

How is this related to $T_4/4^2$?

4. Refine the Picture

Draw the same construction for $n = 8$.



What changes when we move from $n = 4$ to $n = 8$?

What stays the same?

What do you expect the picture to look like for $n = 100$?

Problem

Imagine continuing this process:

$$n = 4, \quad n = 8, \quad n = 100, \quad n = 1000, \quad \dots$$

What geometric object appears to emerge from the rectangles?

Draw it on one of the diagrams above.

5. Let the Spreadsheet Do the Repetition

Open the starter spreadsheet.

The first column contains values of n . Your task is to make the spreadsheet investigate the triangular numbers.

Method A: Build the sequence

Triangular numbers satisfy

$$T_n = T_{n-1} + n.$$

Use this relationship to construct the sequence in the spreadsheet.

What formula did you enter?

What information does the spreadsheet need in order to begin this process?

Method B: Calculate directly

We also know

$$T_n = \frac{n(n+1)}{2}.$$

Use this formula in a new column.

Do the two methods produce the same values?

☐ Yes

☐ No

If they agree, does that prove that the formula is correct for every positive integer n ? Explain.

Normalize

Create another column containing

$$A_n = \frac{T_n}{n^2}.$$

Investigate at least

$$n = 10, \quad 100, \quad 1000.$$

Record your results.

n	A_n
10	
100	
1000	

What value does the spreadsheet suggest A_n is approaching?

How confident are you?

What evidence would make you more confident?

Reflection

The spreadsheet can calculate thousands of examples very quickly.
What can the spreadsheet tell us?

What can it *not* tell us by calculation alone?

6. A New Notation

Instead of writing

$$1 + 2 + 3 + \cdots + n,$$

mathematicians often write

$$\sum_{k=1}^n k.$$

The symbol

$$\Sigma$$

means that we are adding a collection of terms.

So

$$T_n = \sum_{k=1}^n k.$$

Rewrite each expression using sigma notation.

$$1 + 2 + 3 + \cdots + 100$$

$$1^2 + 2^2 + 3^2 + \cdots + n^2$$

$$\frac{1}{n} + \frac{2}{n} + \frac{3}{n} + \cdots + \frac{n}{n}$$

Now return to our rectangle construction.

Each rectangle has width $1/n$, while its height is k/n .

Its area is therefore

$$\frac{k}{n} \frac{1}{n}.$$

The total area of all the rectangles is

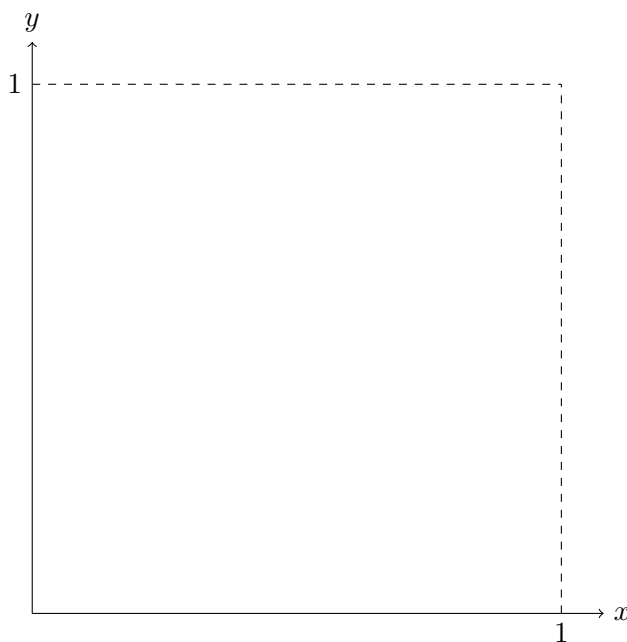
$$\sum_{k=1}^n \frac{k}{n} \frac{1}{n}.$$

7. When Does a Sum Become an Area?

The top edges of our rectangles approach the line

$$y = x.$$

Draw the line $y = x$ and several rectangles below.



As n becomes larger:

- the rectangles become _____;
- the number of rectangles becomes _____;
- their total area appears to approach _____.

We can describe the process symbolically as

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{k}{n} \frac{1}{n} = \frac{1}{2}.$$

Key Idea

The expression on the left is a sequence of *finite sums*.
Every individual calculation uses only finitely many rectangles.
Yet the limiting process appears to determine an exact area:

$$\boxed{\frac{1}{2}}.$$

How can an infinite process produce an exact finite quantity?

One more piece of notation

Modern calculus gives the limiting area another name:

$$\int_0^1 x \, dx = \frac{1}{2}$$

For now, do not worry about calculating integrals using rules or formulas.

Instead, read the symbol as a question:

What quantity is produced when increasingly fine pieces are accumulated?

8. Looking Back / Looking Forward

Return to today's path:

pattern $\longrightarrow$ sum $\longrightarrow$ picture $\longrightarrow$ computation $\longrightarrow$ area.

Choose one transition in this chain.

What new idea became visible when we changed representations?

Historical / Cultural Note

Today we used algebraic formulas, decimal calculations, sigma notation, a spreadsheet, and the modern language of limits to investigate an area.

But the problem is much older than any of these tools.

On Wednesday we will ask how Archimedes could establish exact results about curved figures using only geometry, finite constructions, and carefully controlled approximation.

Exit Question

Suppose someone says:

"A computer can calculate this sum for an enormous value of n , so we do not need the limit."

What would you say in response?